

Mabwell 2MW3311 / CXCR6 Due Diligence Memo

Prepared July 8, 2026. Based on this conversation, the provided Mabwell corporate introduction PPTX, the provided 2MW3311 non-confidential PDF, and public scouting of Mabwell disclosures.

Executive Readout

CXCR6/2MW3311 is a scientifically plausible but still high-risk IBD target/asset. The strongest case is selective depletion of activated, tissue-resident pathogenic T cells rather than another cytokine blockade. The main weakness is translation: no human efficacy data, no clear IBD genetics validation, and material safety questions around depleting CXCR6-positive tissue-resident immune cells.

- Target validity in IBD: moderate and emerging, supported by immune-cell biology and animal efficacy, but not clinically validated.
- Asset position: IND-stage depleting anti-CXCR6 antibody with strong company-presented mouse IBD data and acceptable early NHP tox signals.
- Company credibility: real, listed, operational biotech with commercial products and partnerships, but still loss-making and execution-risky.
- Important discrepancy: source PDF calls the asset 2MW3311, while Mabwell public disclosures refer to 9MW5211 as an undisclosed-target depleting antibody for IBD/MS. Confirm whether these are the same asset or a renamed/code-switched program.

1. CXCR6 Target Validity In IBD

Assessment: moderate / emerging. Plausible mechanism, high translational risk.

- CXCR6 is positioned as a marker of activated, clonally expanded, tissue-homing or tissue-resident T cells, including Th1/Th17 and cytotoxic T-cell subsets.
- The CXCL16-CXCR6 axis plausibly contributes to immune-cell migration, tissue residency, and local inflammatory amplification.
- The asset deck states that pathogenic UC-associated subsets, including Th17, Tc17, CD8 stem-like, and CD8 TGZMK cells, are elevated in active inflamed tissue and highly express CXCR6.
- This is differentiated from cytokine blockade: the therapeutic hypothesis is depletion of disease-driving cells, potentially enabling deeper or more durable remission.
- The highest-risk gap is human translation. Publicly available evidence does not show CXCR6 as a well-established IBD genetics target, and there is no public clinical efficacy data for CXCR6 blockade/depletion in UC or CD.
- Safety biology needs careful testing because CXCR6 is not gut-specific and may be expressed on protective tissue-resident, antiviral, mucosal-surveillance, or regulatory immune populations.

2. 2MW3311 Asset Overview And Key DD Points

What the source deck claims

Dimension	Source-deck summary	DD interpretation
Modality	Humanized anti-CXCR6 monoclonal antibody; specific and potent for human and cynomolgus CXCR6.	Good translational species coverage for tox/PK. Need affinity, epitope, internalization, and receptor occupancy data.

Dimension	Source-deck summary	DD interpretation
Effector function	Defucosylation-enhanced ADCC designed to deplete CXCR6-positive pathogenic T cells.	Potentially differentiating but safety-relevant. Need depth/duration/reversibility of depletion across tissues.
IBD POC	T-cell transfer mouse IBD model: prevented weight loss, depleted CXCR6-positive T cells in blood and mesenteric nodes, lowered colon/spleen weight, improved histology.	Strong preclinical signal, but transfer colitis is not enough for UC/CD confidence. Need orthogonal models and human biopsy PD bridge.
Benchmarking	Anti-CXCR6 compared favorably with upadacitinib and anti-TL1A in mouse pathology/fibrosis readouts.	Interesting, but dosing, route, exposure, and model sensitivity must be normalized before treating this as competitive proof.
Toxicology	4-week repeated-dose SC cynomolgus tox: NOAEL 30 mg/kg; no major abnormalities reported; transient IL-6 increase returned to baseline.	Encouraging early package. Need full GLP report, TK/ADA, histopathology, infection observations, immune subset recovery.
Development	IND approved for IBD in China and US; MS in China. SC high-concentration formulation at 150 mg/mL.	Regulatory entry is credible. Human SAD/MAD design and biopsy/PD endpoints are critical.

Must-answer diligence questions

- Is 2MW3311 identical to public 9MW5211? If yes, document all code-name history, target-disclosure restrictions, IND cross-references, and IP ownership.
- What is the full human translational package: active UC/CD biopsy expression, disease severity correlation, tissue versus blood correlation, and proposed responder biomarker?
- Can Mabwell measure target engagement and depletion without assay interference from the therapeutic antibody?
- What are the consequences of depleting CXCR6-positive immune cells in gut, liver, skin, CNS, lung, and viral challenge contexts?
- What is the dose/exposure margin between pathogenic-cell depletion and broader immune depletion?
- What is the Phase 1/2 design: SAD/MAD, healthy-volunteer versus patient entry, SC dose range, biopsy endpoints, fecal calprotectin/CRP, endoscopy timing, and stopping rules?

3. Mabwell Company Credibility

Assessment: credible operating biotech, but financially and executionally risky.

- Mabwell is publicly listed on SSE and HKEX under 688062.SH and 02493.HK.
- Company materials describe 16 pipeline products, 4 approved/commercialized products, 1 candidate under NDA/BLA review, and 2 candidates in pivotal trials.
- Publicly described capabilities include antibody/ADC/TCE platforms, Taizhou and Shanghai manufacturing sites, commercial products, and overseas regulatory/manufacturing claims.
- External credibility signals include disclosed partnerships/license-outs with DISC Medicine, Calico, Qilu, and Aditum/Kalexo.
- Mabwell public releases on May 8, June 5, and June 24, 2026 describe 9MW5211 as an IBD/MS depleting antibody with FDA and NMPA IBD clearance and NMPA MS clearance.
- Financial risk remains material: Q1 2026 operating income was RMB127.6 million, net loss attributable to shareholders was RMB241.9 million, and operating cash outflow was RMB234.2 million.

- Pipeline breadth is high for a still-loss-making biotech; prioritization, development bandwidth, and funding runway should be assessed before relying on execution timelines.

Decision Framing

Question	Current view	Decision impact
Is the target worth pursuing?	Potentially, if human biopsy and PD data confirm disease-selective pathogenic CXCR6-positive populations.	Proceed to deeper diligence, not immediate conviction.
Is the asset differentiated?	Yes in concept: depleting pathogenic T cells could differentiate from cytokine/pathway modulation.	Differentiation depends on safety margin, biomarkerability, and clinical onset/durability.
Is Mabwell credible?	Yes, as a real listed biotech with products, manufacturing, and partnerships.	Partner risk is manageable but requires financial, CMC, and governance diligence.
Biggest near-term risk	Translation from mouse and expression biology to human IBD efficacy and safety.	Human tissue package and Phase 1/2 biomarker design should drive deal gating.

Sources

- User-provided source: Mabwell_Corporate_Introduction_202606_BD.pptx.
- User-provided source: 2MW3311_CXCR6_BD_Non-Confidential_202606_for_Takeda.pdf.
- Mabwell FDA IND release for 9MW5211: https://www.mabwell.com/en/news_info/id-222.html.
- Mabwell NMPA IBD IND release for 9MW5211: https://www.mabwell.com/en/news_info/id-230.html.
- Mabwell NMPA MS IND release for 9MW5211: https://www.mabwell.com/en/news_info/id-236.html.
- Mabwell R&D; / pipeline page: <https://www.mabwell.com/en/rd.html?yf=z>.
- Mabwell about page: <https://www.mabwell.com/en/about.html>.
- Mabwell 2026 first quarterly report:
https://manager.wisdomir.com/files/770/2026/0428/20260428224501_40951132_en.pdf.
- Subagent scouting summary from this conversation, including public-source checks on CXCR6, 2MW3311, 9MW5211, and Mabwell disclosures.

Note: This memo is a non-confidential diligence synthesis from available materials. It is not a clinical, regulatory, legal, or investment recommendation.